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To cite this article: O. Baussens *et al* 2022 *JINST* **17** C11008

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RECEIVED: August 25, 2022

REVISED: October 3, 2022

ACCEPTED: October 19, 2022

PUBLISHED: November 22, 2022

23RD INTERNATIONAL WORKSHOP ON RADIATION IMAGING DETECTORS
26–30 JUNE 2022
RIVA DEL GARDA, ITALY

Characterization of High-Flux CdZnTe with optimized electrodes for 4th generation synchrotrons

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ABSTRACT: In this collaboration work between IMEM-CNR and the ESRF, we explore the potential of the Redlen Technologies High-Flux Cadmium Zinc Telluride (HF-CZT) material for direct X-ray detection at fluxes ranging from 10^7 to 10^{12} photons $\text{mm}^{-2} \text{s}^{-1}$. HF-CZT single crystals with optimized electroless gold and sputtered platinum electrodes are characterized both under dark conditions and under X-ray irradiation. We report low leakage current under dark conditions (<100 pA/ mm^2 at 5 kV/cm). We also report good linearity ($R^2 > 0.999$) under moderate to high incident X-ray flux (10^7 to 10^{10} photons $\text{mm}^{-2} \text{s}^{-1}$) as well as good stability and reproducibility for incident X-ray fluxes up to 10^{12} photons $\text{mm}^{-2} \text{s}^{-1}$. Finally, we study transient phenomena in the seconds time range and report fast stabilization time (<200 ms) and negligible afterglow in this time range.

KEYWORDS: Instrumentation for synchrotron radiation accelerators; X-ray detectors

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1 Introduction

With the rise of 4th Generation Synchrotron Light Sources such as the Extremely Brilliant Source (EBS) of the ESRF [1], the need for direct X-ray detection under high photon flux ($>10^{+7}$ photons $\text{mm}^{-2} \text{s}^{-1}$) with moderate to high photon energy (30–100 keV range) has increased. Semiconductor materials with high atomic number that can be grown into thick (≥ 1 mm) single crystals are needed for high energy detection. In that respect, Cadmium Zinc Telluride (CdZnTe or CZT) is a good candidate. The CZT single crystals grown with the Traveler Heat Method (THM) can be thicker than 10 mm and have demonstrated high electron transport properties ($\mu\tau_e > 10^{-2} \text{ cm}^2/\text{V}$) [2]. However, their poorer hole transport properties have been an issue at high flux as they lead to radiation-induced polarization phenomena. Recently, Redlen Technologies succeeded in overcoming the issue by enhancing the hole transport properties ($\mu\tau_h > 10^{-4} \text{ cm}^2/\text{V}$) for high-flux applications [2]. This novel high-flux CZT (HF-CZT) seems promising as it limits the polarizing phenomena observed in standard CZT at high photon flux [2, 3]. However, when it comes to electrical contacts, it appears that the standard electroless gold blocking contacts lead to much higher leakage current in HF-CZT than in standard CZT. The IMEM-CNR laboratory (Parma) has been developing novel contacts more suited to the HF-CZT. In this work, Redlen Technologies HF-CZT single crystals with IMEM-CNR optimized electrodes are characterized at the ESRF under both dark conditions and high-flux ($>10^{+7}$ photons $\text{mm}^{-2} \text{s}^{-1}$) X-ray irradiation. It is the first time, to the best of our knowledge, that the Redlen Technologies HF-CZT has been characterized under X-ray fluxes above 10^{+6} photons $\text{mm}^{-2} \text{s}^{-1}$ [4].

2 Methods: HF-CZT detectors

HF-CZT detectors were purchased from Redlen Technologies and re-processed in IMEM-CNR. Two samples, with approximate dimensions $5 \times 5 \text{ mm}^2$ and thickness 1.5 mm, were fabricated with identical electrode layout: planar cathode and pixelated anode (figure 1(a)). The anode layout consists of two single pixels and a 2×2 pixel matrix. All of the pixels are $500 \mu\text{m}$ squares. On the anode side, the two samples have sputtered platinum electrodes. On the cathode side, sample 1 has an electroless gold electrode while sample 2 has a sputtered platinum electrode. The main characteristics of the two samples are summarized in table 1. Both samples were mounted onto PCBs using bicomponent epoxy glue. They were then bounded using copper wires previously soldered onto the PCBs and glued on the pixels with conductive silver epoxy (figure 1(b)). During measurements, the samples were placed in a custom made aluminium alloy shielding box (figure 1(c)). The shielding box provided dark conditions, its entrance window was in opaque kapton tape. The temperature inside the shielding box was controlled using a Julabo FL300 recirculating cooler and measured with a Pt100 probe. To prevent water condensation, a nitrogen flow was maintained inside the shielding box. All characterizations discussed in this paper were performed on single pixel 1 (pixel on the top left hand corner in figure 1(a)). The cathode of the sample under test was negatively biased using a ISEG SHQ 224 high voltage power supply and its current was measured using a Keithley 6485 picoammeter.

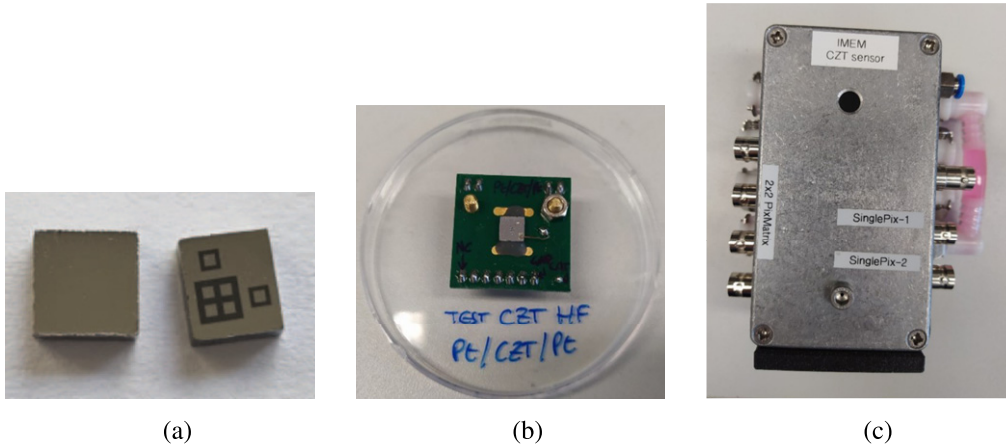


Figure 1. (a) On the left, the planar cathode. On the right, the pixelated anode layout. (b) Sample 2 mounted onto its PCB. (c) Shielding box.

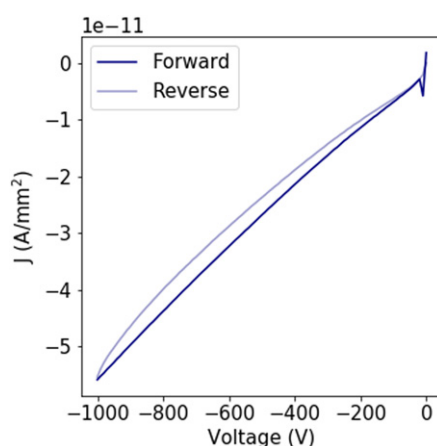
In the interest of improved readability, and because the results of sample 1 and 2 were equivalent for all the characterizations carried out, only the results of sample 2 will be presented throughout the paper. The results of sample 1 are available in the supplementary material and will be discussed alongside the results of sample 2 in the relevant sections of this paper.

Table 1. Characteristics summary of the HF-CZT detectors.

	Cathode contact	Anode contact	Geometry (mm ²)	Thickness (mm)	Pixel size (mm ²)
Sample 1	Au	Pt	5 × 5	1.5	0.5 × 0.5
Sample 2	Pt	Pt	5 × 5	1.5	0.5 × 0.5

3 Dark conditions measurements

The I - V characteristics of both samples were measured at 20 °C under dark conditions between 0 and -1000 V using -10 V steps. Figure 2 displays the I - V characteristic of sample 2. Figure S1 (supplementary material) displays the I - V characteristic of sample 1. For both samples, an unexpected peak of current is visible at low bias on the forward scan of the I - V characteristics. This peak is believed to be related to the inaccuracy of the instrumentation at low voltage. It can also be noted that the I - V characteristics of both samples present some hysteresis but this phenomenon is very limited with the deviation between the forward and reverse scans being less than 10%. The dark current densities at -1000 V (-5 kV/cm) of sample 1 and 2 (Au/CZT/Pt and Pt/CZT/Pt) are 90 pA/mm² and 60 pA/mm² respectively. These dark current density values constitute a 1 order of magnitude improvement compared with the standard Redlen Technologies pixelated HF-CZT (2 mm thick) [3] and a 4 orders of magnitude improvement compared with other electrode configurations (Au/CZT/Au and Pt/CZT/Au) previously tested on HF-CZT at IMEM-CNR. These comparisons underline the role of the platinum anode in the leakage current reduction. To conclude, the dark current results confirm that the electrodes optimized by IMEM-CNR are successful blocking contacts.

**Figure 2.** I - V characteristic of sample 2 at 20 °C under dark conditions.

4 Behaviour under X-ray irradiation

4.1 Moderate to high X-ray flux at the BM05 beamline

Both samples were characterized under moderate to high X-ray flux (3×10^7 to 8×10^9 photons $\text{mm}^{-2} \text{s}^{-1}$) at the BM05 beamline of the ESRF using a 20 keV monochromatic synchrotron beam. The beam was collimated to the pixel size ($500 \mu\text{m} \times 500 \mu\text{m}$). The incident flux was adjusted using aluminium attenuators and measured with a $500 \mu\text{m}$ silicon diode.

Three measurements were performed on each samples using the setup described: first, a pulsed irradiation measurement, then, an I - V characteristic measurement, and finally, a long term stability measurement at a given bias and flux. The objective was to verify the stability and the linearity of the samples with the incident X-ray flux.

The pulsed irradiation measurement consisted in submitting the sample under test, biased at -1000 V , to increasing X-ray fluxes. Seven X-ray fluxes within the flux range were tested. Each flux was applied for 10 minutes. Between flux steps, the beam was cut off for 5 minutes using the maximum thickness of Al attenuators available (11.5 mm). The current of the sample under test was recorded every 200 ms for the entire duration of the test. Figure 3 shows the pulsed irradiation characterization of sample 2 as well as its linearity curve with incident flux. The results of sample 1 are available in figure S2 of the supplementary material. Both samples show good stability and good linearity for all the fluxes tested.

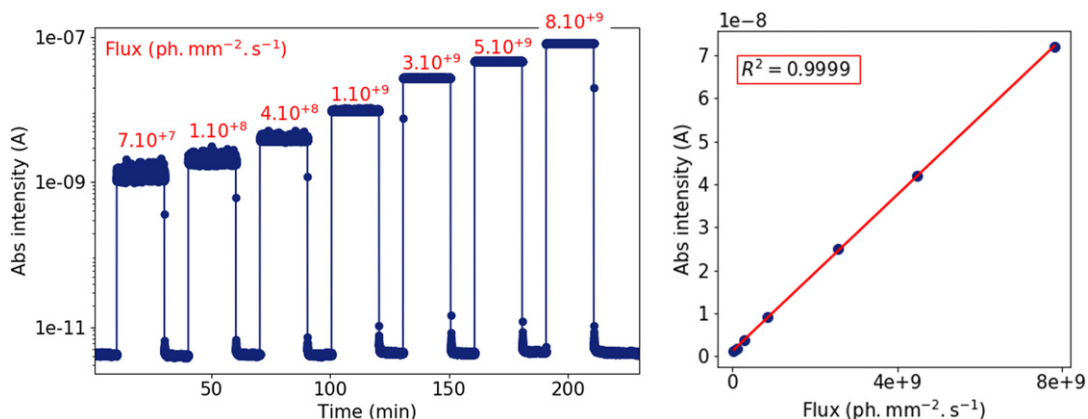


Figure 3. On the left, 20 keV pulsed irradiation of sample 2 measured on the BM05 beamline. In red, above each pulse, the corresponding incident flux in photons per millimeter squared per second. On the right, the linearity curve of sample 2 with incident X-ray flux.

For the I - V characteristics and the time stability measurements three incident fluxes were selected. The results of these measurements for sample 2 are shown in figure 4. For sample 1, they are available in the supplementary material, figure S3. The I - V characteristics were measured for biases between 0 and -1000 V using -5 V steps. For both samples and for all fluxes tested, the current reaches the saturation value corresponding to complete charge collection above -100 V bias. It is to be noted that, to improve readability, figure 4 only displays the forward scans of the

I - V characteristics. The reverse scans show no significant deviation from their forward counterparts. For the time stability measurements, the samples were biased at -1000 V and their current was measured every 5 s for 80 min. Both samples showed good time stability regardless of the incident flux.

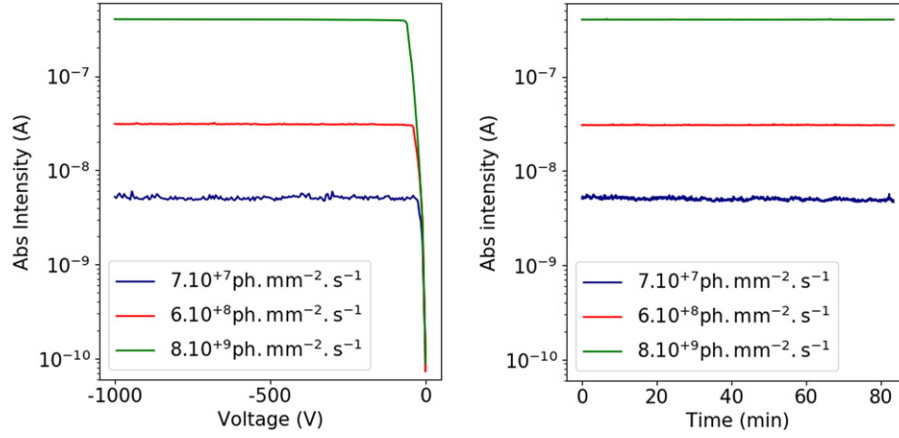


Figure 4. On the left, I - V characteristics of sample 2 under X-ray irradiation. On the right, time stability of sample 2 under X-ray irradiation.

The characterizations performed on the BM05 beamline of the ESRF demonstrated that the samples are stable and linear under incident X-ray fluxes ranging from $3 \times 10^{+7}$ to $8 \times 10^{+9}$ photons $\text{mm}^{-2} \text{s}^{-1}$. However, ongoing detector developments of the ESRF require sensors capable of sustaining fluxes up to 10^{+10} photons $\text{mm}^{-2} \text{s}^{-1}$. This flux cannot be reached with the current setup available at the BM05 beamline. For this reason, another set of characterizations was carried out at the ID19 beamline of the ESRF.

4.2 High to very high X-ray flux at the ID19 beamline

The ID19 beamline of the ESRF is capable of reaching fluxes several orders of magnitude beyond 10^{+10} photons $\text{mm}^{-2} \text{s}^{-1}$. However, we could only access the beamline for a very limited amount of time and thus only platinum electrodes (sample 2) could be tested. Two characterizations were performed: pulsed irradiation measurements and I - V characteristic measurements. Both characterizations were performed using a 19 keV 1% monochromatic synchrotron beam produced by the U32 insertion device of the beamline [5].

The pulsed irradiation measurements were performed for four fluxes ranging between $2 \times 10^{+9}$ and $1 \times 10^{+12}$ photons $\text{mm}^{-2} \text{s}^{-1}$. The incident flux was adjusted by varying the gap of the undulator. For each of the four fluxes, sample 2 was irradiated during 10 s. Between flux steps, the beam was cut off for 15 s. The pulsed irradiation measurement was repeated three times and was found to be reproducible. Figure 5 shows one pulsed irradiation characterization of sample 2. The current density of sample 2 is stable for all the incidents fluxes tested.

Please note that the four incident fluxes were not measured but were estimated from the sample current, hypothesizing complete charge collection above -500 V. They were also estimated using

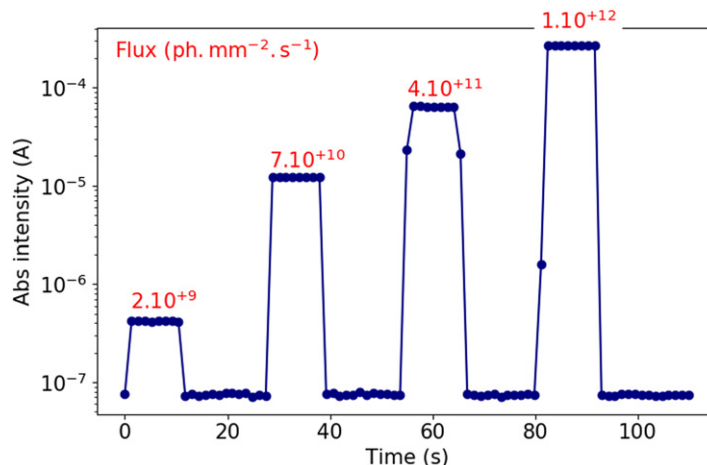


Figure 5. Pulsed irradiation of sample 2 at 19 keV on the ID19 beamline. In red, above each pulse, the corresponding incident flux in photons per millimeter squared per second.

the beamline setup calculation software. Both estimations were consistent. However, since these are estimations, the flux values given in figures 5 and 6 only illustrate the orders of magnitude involved and cannot be used to plot the linearity of the sample's current with incident X-ray flux.

The I - V characteristics were measured between 0 and -1000 V with -10 V steps for the same four fluxes used during the pulsed irradiation measurements (figure 6). The characteristics show stable and reproducible behaviour of both forward and reverse scans for fluxes up to 10^{+11} photons $\text{mm}^{-2} \text{s}^{-1}$. Above this flux, the I - V characteristics start getting deformed and a hysteresis phenomenon appears for the highest flux tested. Possible explanations for these effects include a complex time-evolution of the electric field in the sample due to electron trapping and irradiation-induced polarization [6]. The study of these effects is part of ongoing work and will not be discussed further in this paper.

To summarize, the characterizations performed on the ID19 beamline of the ESRF demonstrated a stable and reproducible behaviour of sample 2 under 10 s X-ray irradiation with incident fluxes up to 10^{+12} photons $\text{mm}^{-2} \text{s}^{-1}$. However, an hysteresis effect starts to appear on the I - V characteristics above 10^{+11} photons $\text{mm}^{-2} \text{s}^{-1}$.

4.3 Afterglow: preliminary results

Afterglow (or residual photocurrent) measurements were not specifically performed on the samples. However, it is possible to make observations about afterglow from the pulsed irradiation results of figure 3. Because the frequency used to measure the current during the pulsed irradiation characterization was 5 Hz, the analysis of the afterglow can only be done for time constants above 200 ms. Lower time constants will be investigated in future works.

From figure 3, three observations can be made regardless of the incident X-ray flux. Firstly, the rise time for all pulses is below or equivalent to the measurement limit of 200 ms. Secondly, the photocurrent is stable throughout irradiation, no transient phenomena that could be consistent with significant trapping mechanisms are observed. And thirdly, after the irradiation is stopped, it

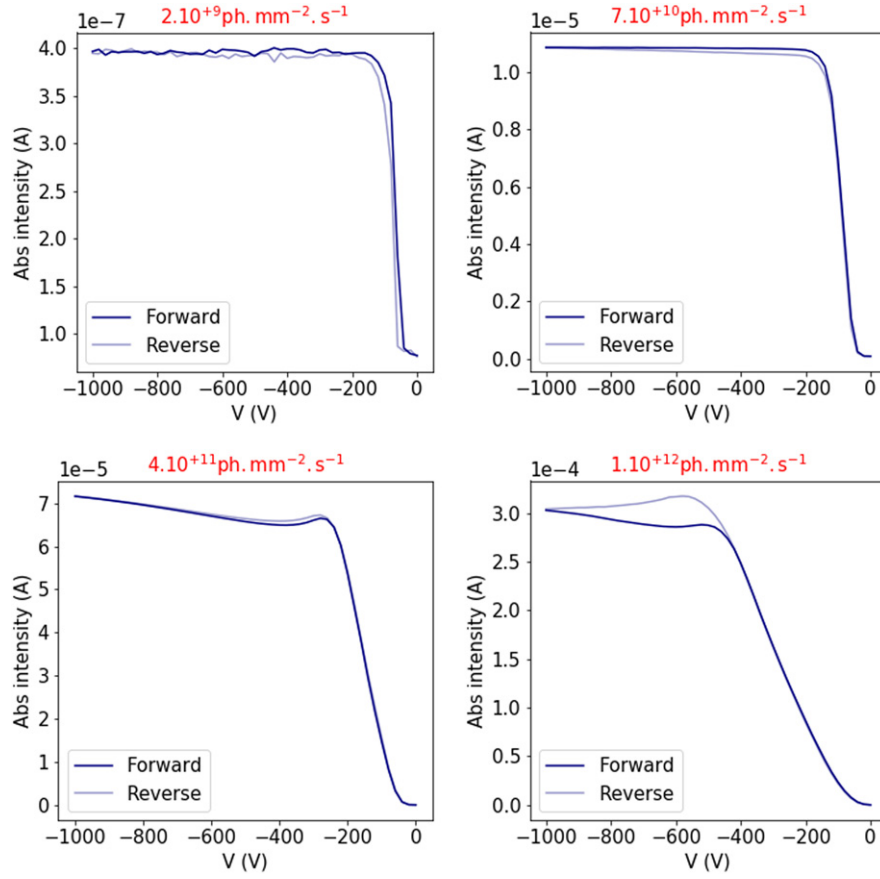


Figure 6. I - V characteristics of sample 2. In red, above each plot, the corresponding incident X-ray flux.

takes 400 ms (2 measurement points) for the current to fall below dark current levels (10^{-11} A). Furthermore, figure 7 shows the afterglow for each pulse 1 s after the X-ray irradiation is stopped and compares them with the photocurrent values during irradiation. The afterglow measured is below 0.5% of the photocurrent level measured during irradiation regardless of the incident X-ray flux. The first two observations, demonstrate that the HF-CZT has fast stabilization times (<200 ms) and is not affected by polarization effects for the flux range tested. The third observation and the results of figure 7 demonstrate that, for applications with frame rates in the seconds time range, the afterglow of HF-CZT is negligible even after minutes long irradiation under a 10^{+10} photons $\text{mm}^{-2} \text{s}^{-1}$ incident X-ray flux. These conclusions are consistent with recent results presented in the IEEE NSS/MIC/RTSD 2021 conference [7].

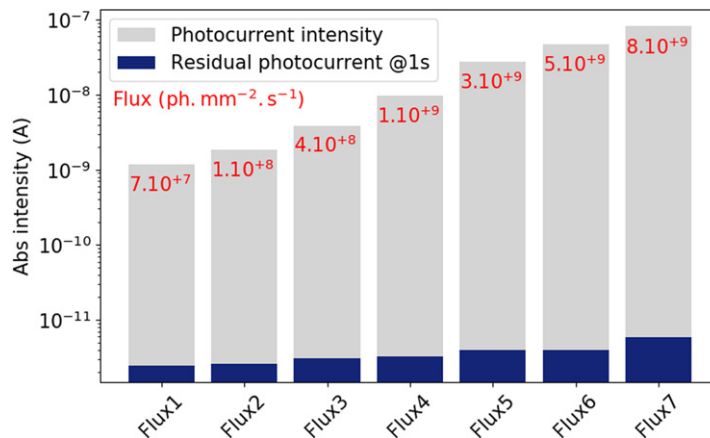


Figure 7. Comparison of the residual current 1 s after irradiation has stopped with the previous photocurrent.

5 Conclusion

The aim of this work was to characterize the high-flux CZT material from Redlen Technologies under high X-ray fluxes ranging from 10^{+7} to 10^{+12} photons $\text{mm}^{-2} \text{s}^{-1}$. Two devices with optimized electrode configurations (Au/CZT/Pt and Pt/CZT/Pt) were fabricated in IMEM-CNR from the HF-CZT material. Both of these optimized electrode configurations demonstrated significantly reduced leakage current ($<100 \text{ pA/mm}^2$ at -5 kV/cm and at 20°C) compared with standard Redlen Technologies pixelated platinum electrode configuration. The linearity, stability and repeatability of HF-CZT sensors under incident X-ray flux was demonstrated for both electrode configurations between 10^{+7} to 10^{+10} photons $\text{mm}^{-2} \text{s}^{-1}$. The Pt/CZT/Pt configuration was also measured under even higher incident X-ray fluxes. From these measurements, the stability and reproducibility of HF-CZT were demonstrated for X-ray fluxes up to 10^{+12} photons $\text{mm}^{-2} \text{s}^{-1}$ though a hysteresis effect was observed on the I - V characteristics above 10^{+11} photons $\text{mm}^{-2} \text{s}^{-1}$. Finally, the afterglow and stabilization time of HF-CZT in the seconds time range were studied for both electrode configurations. In this time range, the afterglow was shown to be negligible and the stabilization time was shown to be below 200 ms. To conclude, the potential of Redlen Technologies HF-CZT material for the detection of high-flux X-ray was demonstrated. Moreover, both configurations, with Au or Pt cathodes, were shown to be stable and successful blocking contacts, leaving the choice of a particular configuration to purely technological considerations (deposition process availability, cost, time constraints. . .). Future work will be dedicated to the study of the hysteresis effect observed at very high X-ray incident flux as well as to a more in depth study of the afterglow.

Acknowledgments

The authors would like to thank Phil Cook from the BM05 beamline and Bratislav Lukic from the ID19 beamline for their help and technical support.

References

- [1] P. Raimondi, *ESRF-EBS: the extremely brilliant source project*, *Synchrotr. Radiat. News* **29** (2016) 8.
- [2] B. Thomas et al., *Characterisation of Redlen high-flux CdZnTe*, *2017 JINST* **12** C12045.
- [3] S. Tsigaridas, C. Ponchut, S. Zanettini and A. Zappettini, *Characterisation of pixelated CdZnTe sensors using MAXIPIX*, *2019 JINST* **14** C12009.
- [4] M. Charles Veale et al., *Characterization of the uniformity of high-flux CdZnTe material*, *Sensors* **20** (2020) 2747.
- [5] ESRF, *ID19 Beamline description—insertion devices* (2017) Last Modified: Mon 09 January 2017 08:29:22 GMT.
- [6] D.S. Bale and C. Szeles, *Nature of polarization in wide-bandgap semiconductor detectors under high-flux irradiation: application to semi-insulating $\text{Cd}_{1-x}\text{Zn}_x\text{Te}$* , *Phys. Rev. B* **77** (2008) 035205.
- [7] D. Greiffenberg, M. Andrae, R. Barten and J. Zhang, *High-Z Sensors for Synchrotron Sources and FELs* (2021).